

Implementation of Secure IAM Access Management and EC2 IAM Role Configuration for Amazon S3 Read-Only Access

Objective

The objective of this assignment was to understand and implement Identity and Access Management (IAM) best practices within Amazon Web Services (AWS). The lab focused on creating IAM users, organizing permissions using IAM Groups, assigning AWS Managed Policies, creating an IAM Role for EC2, and securely attaching the role to an EC2 instance. This approach demonstrates the principle of least privilege while enabling secure access to Amazon S3 resources without exposing long-term credentials.

Technologies Used

- Amazon Web Services (AWS)
- Identity and Access Management (IAM)
- Amazon EC2
- Amazon S3
- IAM Groups
- IAM Roles
- AWS Managed Policies
- AWS Management Console

AWS Region Used:

- Asia Pacific (Mumbai)

Assignment Overview

The implementation consisted of the following major tasks:

1. Creation of IAM Users
2. Creation of an IAM Group
3. Assignment of AmazonS3ReadOnlyAccess Policy
4. Addition of IAM Users into the Group
5. Creation of an EC2 IAM Role
6. Assignment of AmazonS3ReadOnlyAccess to the Role
7. Attachment of IAM Role to EC2 Instance
8. Verification of Successful Role Attachment

Step 1 – IAM User Creation

Two IAM users named **alice** and **bob** were created.

Why?

IAM users represent individual identities within AWS.

Creating separate users avoids sharing the root account.

Benefits

- Individual accountability
- Separate authentication
- Better auditing
- Improved security

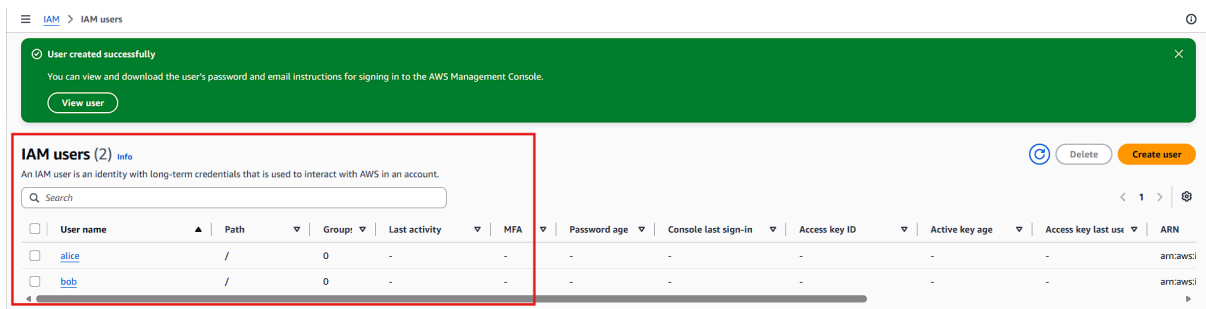


Figure 1: IAM Users Created (alice and bob)

Step 2 – Creating IAM User Group

A user group named **S3_ReadOnly_Group** was created.

Why?

Instead of assigning permissions individually, AWS recommends assigning permissions through groups.

Benefits

- Centralized permission management
- Easier administration
- Reduced configuration errors
- Scalable identity management

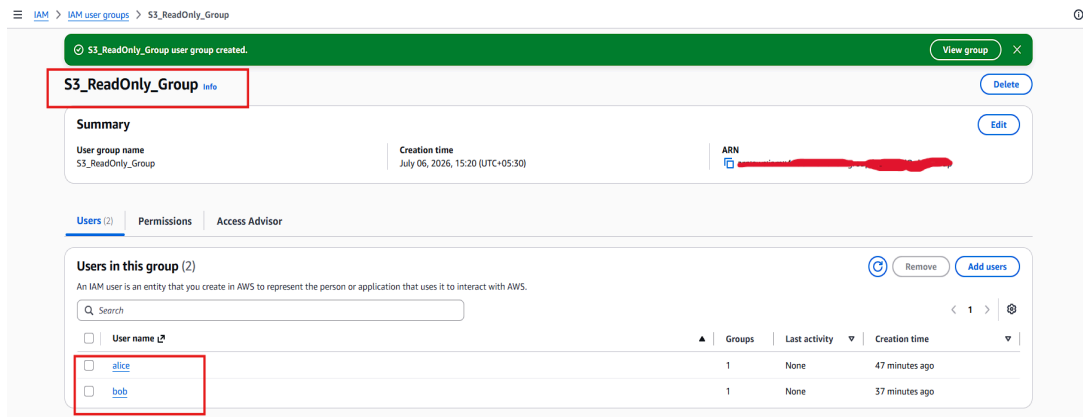


Figure 2: IAM User Group (S3_ReadOnly_Group) Created

Step 3 – Attaching AmazonS3ReadOnlyAccess Policy

The AWS Managed Policy **AmazonS3ReadOnlyAccess** was attached to the IAM Group.

Why?

Users only require read access.

Using AWS Managed Policies simplifies administration.

Benefits

- Least Privilege
- AWS maintained policy
- Automatic policy updates
- Reduced administrative effort

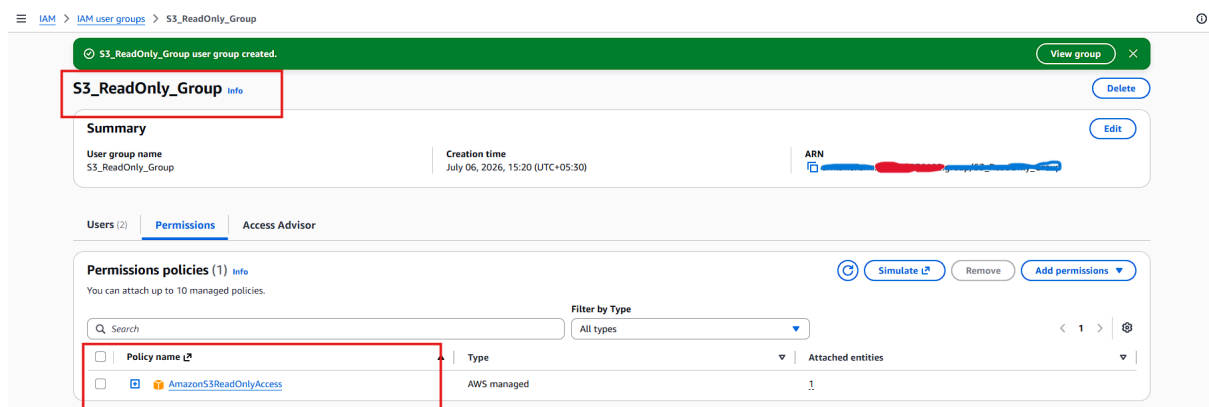


Figure 3: AmazonS3ReadOnlyAccess Policy Attached to IAM Group

Step 4 – Adding Users to the Group

Both users were added into the S3_ReadOnly_Group.

- alice
- bob

Why?

Users inherit permissions automatically from the group.

Benefits

- Easier onboarding
- Simplified access control
- Consistent permissions

Refer to Figure 2, which confirms both IAM users (alice and bob) were successfully added to the S3_ReadOnly_Group.

Step 5 – Creating IAM Role for EC2

An IAM Role named **EC2_S3_Role** was created.

The trusted entity selected was **Amazon EC2**

Why?

Applications running on EC2 should never use permanent AWS credentials.

IAM Roles provide temporary credentials.

Benefits

- Temporary credentials
- Automatic credential rotation
- No hardcoded secrets
- Improved security

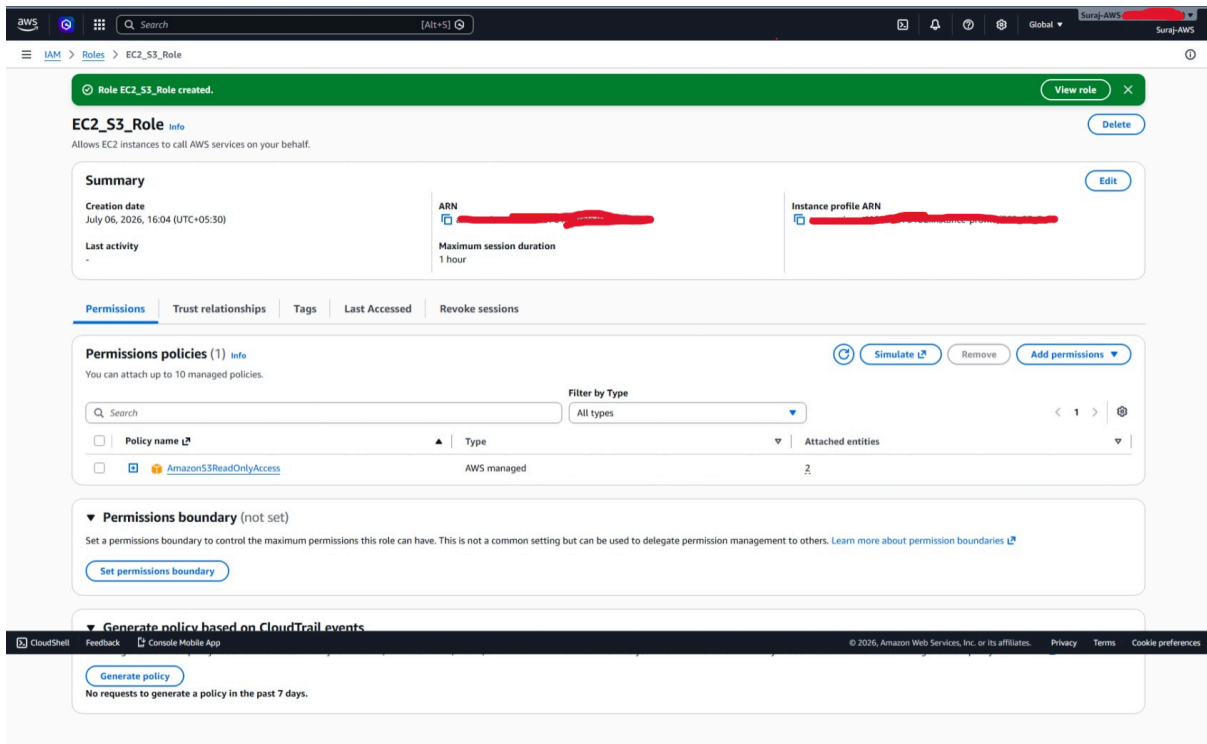


Figure 4: EC2 IAM Role Successfully Created

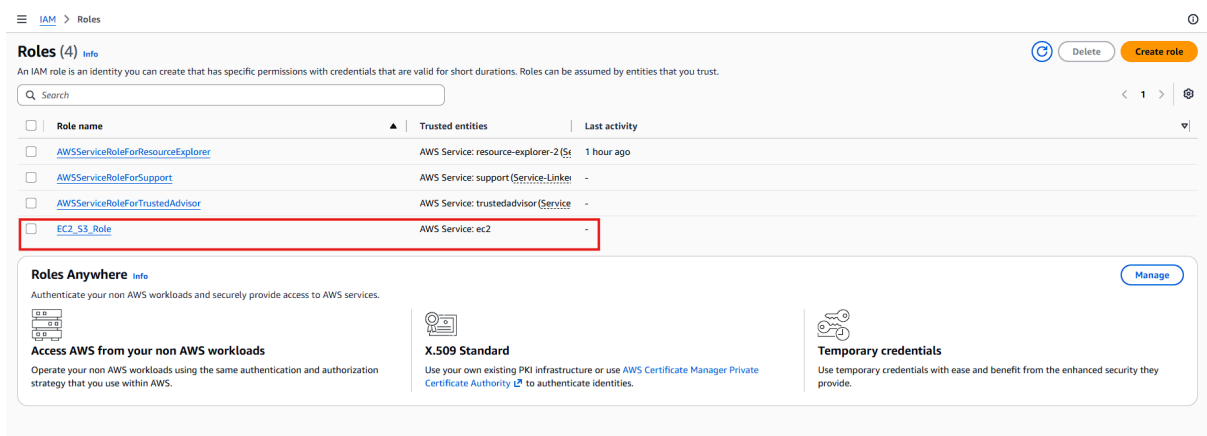


Figure 5: IAM Role Listed Under AWS IAM Roles

Step 6 – Assigning AmazonS3ReadOnlyAccess to the Role

The EC2 role received the **AmazonS3ReadOnlyAccess** policy.

Why?

The EC2 instance only needs to read S3 objects.

Benefits

- Principle of Least Privilege
- Reduced attack surface
- Prevents accidental deletion of S3 objects

Figure 4 confirms that the AmazonS3ReadOnlyAccess policy was successfully attached to the IAM Role.

Step 7 – Attaching IAM Role to EC2

The IAM Role **EC2_S3_Role** was attached to the EC2 instance.

Verification confirmed successful attachment.

Why?

EC2 automatically receives temporary credentials.

No Access Keys are stored on the instance.

Benefits

- Secure authentication
- Automatic credential rotation
- Compliance with AWS security best practices

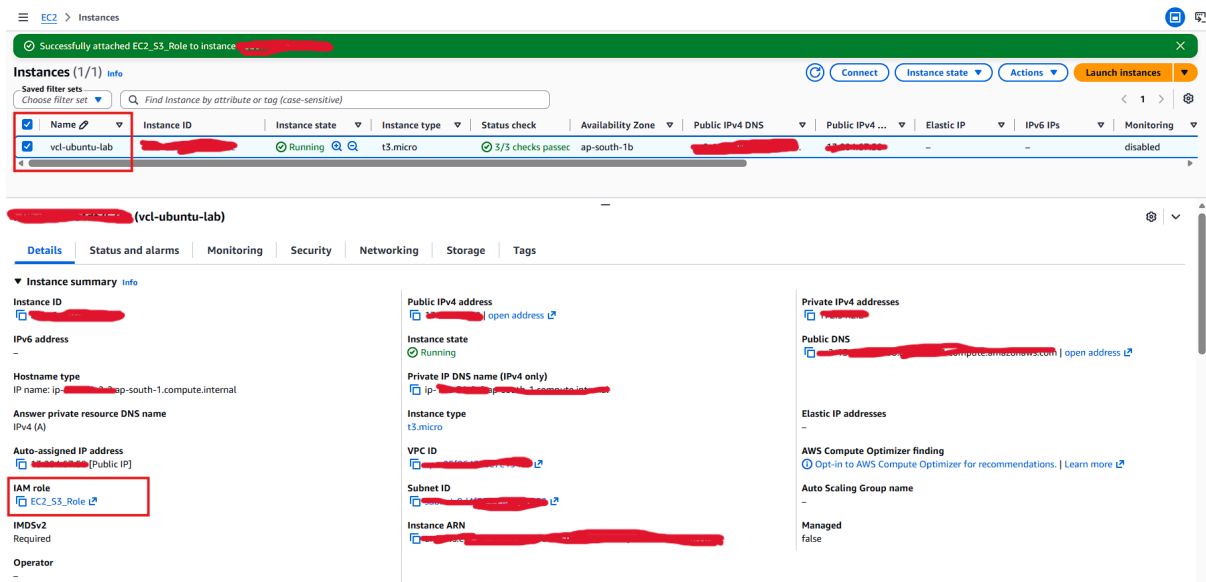


Figure 6: EC2 Instance with Attached IAM Role

Step 8 – Verification

The EC2 Details page confirmed:

IAM Role

EC2_S3_Role

This verified successful role attachment.

Figure 6 verifies that the EC2 instance is successfully associated with the EC2_S3_Role IAM Role.

Security Benefits

The implementation demonstrates several AWS security best practices:

- Principle of Least Privilege
- Role-Based Access Control (RBAC)
- Temporary Credentials
- IAM Group-Based Permission Management
- AWS Managed Policies
- Reduced Credential Exposure
- Centralized Access Management
- Secure Cloud Identity Management

Security Recommendations

For production environments, additional security measures should include:

- Enable Multi-Factor Authentication (MFA) for all IAM users.
- Avoid using the root account for daily operations.
- Apply least privilege by granting only the minimum required permissions.
- Regularly review IAM permissions and remove unused accounts.
- Rotate credentials where long-term credentials are unavoidable.
- Enable AWS CloudTrail for auditing.
- Enable Amazon GuardDuty for threat detection.
- Use AWS Config to monitor configuration drift.
- Enable S3 server-side encryption.
- Block all public access to S3 buckets unless explicitly required.
- Use VPC endpoints for private access to S3 where appropriate.

What NOT to Do & Implications

Action	Implication
✗ Do not use the Root Account	Compromise of the root account grants unrestricted access to the entire AWS environment.
✗ Do not hardcode AWS Access Keys in EC2	Exposed credentials can lead to unauthorized access, data theft, and privilege escalation.
✗ Do not assign AdministratorAccess unnecessarily	Excessive permissions violate the principle of least privilege and increase the impact of compromised credentials.
✗ Do not assign permissions directly to individual users	Inconsistent access control, difficult audits, and increased administrative overhead.
✗ Do not make S3 buckets publicly accessible unless required	Sensitive data may become publicly exposed.
✗ Do not leave unused IAM users active	Dormant accounts can become targets for attackers.
✗ Do not expose AWS Account IDs, ARNs, Instance IDs, or IP addresses in public repositories	Infrastructure details can aid reconnaissance and targeted attacks.
✗ Do not keep EC2 instances running unnecessarily	Increased operational costs and a larger attack surface.
✗ Do not ignore logging and monitoring	Security incidents may go undetected, delaying response and forensic analysis.

Conclusion

This assignment successfully demonstrated the implementation of AWS Identity and Access Management using security best practices. IAM users were organized into a group with the AWS-managed **AmazonS3ReadOnlyAccess** policy, reducing administrative overhead and enforcing consistent permissions. An EC2 IAM role was created and attached to the instance, allowing secure access to Amazon S3 through temporary credentials instead of long-term access keys. The completed configuration aligns with core cloud security principles such as least privilege, role-based access control, and credential minimization, providing a secure and scalable foundation for managing AWS resources.